

Corning ONE™ SD-LAN

Technical Publication – Northbound Interface API (JSON REST)



Overview |

Corning has provided access to a RESTful Northbound Interface for retrieving information from the SD-LAN orchestration platform. Although the SD-LAN system provides a robust and feature rich graphical user interface (GUI), external systems can poll the SD-LAN system to retrieve summary and detailed information from REST APIs.

- Summary system Information including exceptions list
- System wide Switching Device Report
- Endpoint Set Device Report
- Detailed Individual Device Report by Endpoint Name
- Slice View by Endpoint Name and/or Port

The purpose of this document is to provide details regarding the information provided in these APIs.

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Corning ONE™ SD-LAN

Summary System Information – Status Query |

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The Status Query provides a system summary to identify any system anomalies that require network administrator intervention. It is expected that the monitoring system polls the Corning system at least once every ten minutes. Once notified, the operations support personnel login to the to proceed with troubleshooting the issues that are reported.

The following table shows the information returned from the query pointing towards “<site URL>/api/request” either via HTTPS or via POST. It is recommended to use POST so that username/password information is protected.

<https://nnn.nnn.nnn.nnn/api/request?username=admin&password=admin&query=status>

FIELD	Definition	Contents
ftp_active	Number of Optical Edge Devices	<Integer>
switches	Number of Switching Devices	<Integer>
vnetc_ifaces	List of MAC Interfaces for the Orchestration platform VM	<name><mac address>
ftp_lost_contact	Count of Optical Edge Devices out of contact	<Integer>
uptime	Orchestration platform uptime in Seconds	<Floating point>
tenant_licenses	System License Table	
Tenant count		
annotations	Strings inserted in license file	<json>
support	Technical Support Email	<string>
licensing	Sales Support Email	<string>
valid	License file is valid	<TRUE FALSE>
licensed_mac	MAC address embedded in license file	<mac address>
endpoint_use	Count of licensed Endpoints	<integer>
not_after	Expiration Date/Time	<Epoch Time of Day>
missing	License File Existence	<TRUE FALSE>
fqdn	Full Qualified Domain Name	<URL>
valid_hardware	Valid Mac address in license file	<TRUE FALSE>
not_before	License Start Date/Time	<Epoch Time of Day>
voip_limit	Count of licensed POTs ports	<Integer>
version	License File Format	<integer>
voip_use	Number of provisioned voice lines	<integer>
sd_wan_sites	Licensed SD-WAN sites	<integer>
ont_add_on	Site option for RG Features	<TRUE FALSE>
sha256	HMAC	<key>
expired	License file has expired	<TRUE FALSE>
endpoint_limit	Number of licensed endpoints	<integer>
ports_use	Number of provisioned ports	<integer>

last_pg_archive	Timestamp of most recent POSTGRES backup	<Epoch Time of Day>
hostname	Site FQDN	<xxxx.yyyy.net>
sites	Number of SD-WAN sites	<integer>
site_vfs_status	List of SD-WAN site status	<json>
fttp_rebooted	Number of optical edge devices rebooting in last 24 hours	
site_freespace	Hypervisor free space for Edge deployment scenario	<integer>
bufferwatch	Management interface monitor for directly connected deployment scenario "Starting", "Normal", "Time to live exceeded", "Network is unreachable", "Destination Host Unreachable", etc	<string>
installations	Number of SD-WAN networks	<integer>
timestamp	Time of Query	<Epoch Time of Day>
problems	List of active problems Each problem entry may have these fields:	
installation_name	SD-WAN installation name for Edge deployment scenario	<text>
tenant_id	SD-WAN Tenant number	<integer>
element	Domain-Site-Endpoint	<text>.<text>.<text>
tenant_name	Name of SD-WAN Tenant	<text>
message	Switch Error Info	<text>
element_id	Internal object number	<integer>
element_name	Chassis ID (hostname, serial or MAC address)	<text>
Installation_id	SD-WAN installation number	<integer>
date	Human readable UTC date	<text>
freespace	Disk free space by file system	<percent available>
switch_status		
Network		
endpoint_name	Dotted path of Domain, site and endpoint name	<domain.site.endpoint>
.state	Communication State	True/False
Temp	Array of Temp Sensors	Degrees Celsius
PSU	Power supply status - "OK", "BAD", "ABSENT" (or other hardware vendor specific)	<string>
package	vNETC Software package (e.g. : "5.4.8.157",)	<string>

site_wan_status	Not used for SD-LAN	
loadavg	Last 5 minute sample of vNETC load average calculation	<floating point>
tenants	Number of tenants, should always be 1. Use of multiple tenants is deprecated	<integer>
user_email	Connecting user's username	<string>

Systemwide Switching Device Report - Networks |

The Networks Query provides a system wide switching device report. It provides detailed information of each connected device, organized by Domain and Site.

The following table shows the information returned from the query pointing towards “<site URL>/api/request” either via HTTPS or via POST. It is recommended to use POST so that username/password information is protected.

<https://nnn.nnn.nnn.nnn/api/request?username=admin&password=admin&query=networks&setname=xxxx&site=yyyy>

“setname” can be used to limit the results to just the named endpoint set.

“site” can be used to limit the results to just the named site.

FIELD	Definition	Contents
Domain	Name of domain	<string>
Site	Name of Site	<string>
dev_id	Hardware Device ID – internal database identification of a hardware device	<integer>
hw_mfg	Device Manufacturer	<string>
endpoint_id	Endpoint ID – internal database identification of a service endpoint. Provisioning is tied to Endpoint, and Endpoint is tied to hardware device	<integer>
endpoint_name	Name of service endpoint – usually a physical location/meaningful location	<string>
hw_ip	Assigned IP Address	<nnn.nnn.nnn.nnn>
topology_depth	Number representing the tier depth of the device. (eg. TOR devices are considered Tier 1 and then aggregation devices directly connected to the TOR are Tier2)	<integer>

connections	List of all detected connections (Active [Up] / Inactive [Down])	
Port Number	Port Number – local port number. On entry for each port connected to managed device	<json>
remote_port	Port number of connected device	<integer>
remote_dev_id	Device ID of connected device	<integer>
mode	Connection Mode - "up" (service ports from LAN towards WAN connection), "down" (connection to other switches/devices), "cross" (connections between redundant pairs of switches)	<string>
local_port_name	Manufacturer specific port name of this port	<string>
remote_name	Hostname of connected device	<string>
remote_port_name	Manufacturer specific port name of remote port	<string>
is_tor	Indication that this switch is at the top of the LAN connected to the WAN	<True False>
hw_ident	MAC address or serial number identifying the chassis – provided in SNMP and LLDP information	<string>
lost_contact	Time that contact was lost between device and orchestration platform. Null if device is in communication	<string>
endpoint_id	Associated Endpoint ID	<integer>
hw_model	Manufacturer Hardware Model	<string>
attributes		<json>
Fan1	Device Fan Status – “OK”, “FAIL”, “Missing”	<string>
PSU1	Device Power Supply Unit Status– “OK”, “FAIL”, “Missing”	
Temp1	Device Sensor Temperature in Celsius	<float>
mgmt_vlan	Management VLAN assigned	<string>
ActiveImage	HW Manufacturers Image/software load	<string>
hw_hostname	Hostname of device	<string>
hw_sn	Hardware Serial Number	<string>



The Detailed Individual Device Report adds the following information to the switch information provided within the System wide Switching Device Report

- 1) MAC addresses – indicating VLAN and any available aging information per port
- 2) Statistics – all historical counters/sensor readings for the device as well as per port statistics/readings.

The statistics period is defined by start and end time parameters in the query.

The following table shows the additional information (added to the information provided by the “networks” query) returned from the query pointing towards “<site URL>/api/request” either via HTTPS or via POST. It is recommended to use POST so that username/password information is protected.

https://nnn.nnn.nnn.nnn/api/request?username=admin&password=admin&query=device_details&dev_id=dddd&endpoint_id=eeee&endpoint_name=xxxx&start=nnnnnnnnn&stop=nnnnnnnnn

“dddd” is an integer dev_id. “eeee” is an integer endpoint_id. “xxxx” is a text endpoint name. Exactly one of these must be provided. The dev_id is slightly more efficient so is preferred if available.

The start and stop times are monotonic UTC epoch timestamps. The sample step (sample “bucket” size) depends on the start time. Start times less than 8 days ago have sample durations of 5 minutes, and less than 35 days ago have sample durations of 30 minutes.

Device Wide		
dev_id	Device ID of this device	<integer>
endpoint_name	Name of Endpoint	<string>
hw_ident	MAC address or serial number identifying the chassis – provided in SNMP and LLDP information	<string>
hw_mfg	Device Manufacturer	<string>
hw_model	Hardware Model	<string>
hw_sn	Hardware Serial number	<string>
lost_contact	Time that contact was lost between device and orchestration platform. Null if device is in communication	<integer>

lost_reason	reason the connection was lost if known. (eg. "No contact within xxx", "System restart", "Management connection lost")	<string>
management_ip	IP address used to communicate with the management interface of the device	<string>
statistics		
step	Length of each Time period in seconds	<integer>
start	UTC Timestamp of first reading	<integer>
stop	UTC Timestamp of last (most recent) reading	<integer>
device	JSON object of dotted paths. Each path lists the sample values	<JSON>
ports	JSON object of port names	<JSON>
port	JSON object of dotted paths. Each path lists the sample values	<JSON>
ports	JSON object of port names. Each port records the following:	
port_status	Service status (eg. Up, Down, etc.)	<string>
last_link_change	UTC Timestamp of last port_status change	<integer>
poe_status	PoE delivery status (eg. Powered, unpowered, not provisioned)	<string>
sfp_type	Copper or Optical	<string>
rx_optical_level	If Optical, dBm level	<float>
tx_optical_level	If Optical, dBm level	<float>
sfp_vendor_name		<string>
sfp_vendor_part_number		<string>
sfp_vendor_serial_number		<string>
mac_addresses	JSON object of port names. Each port records the following:	JSON
mac	Mac address in colon separated format – nn:nn:nn:nn:nn:nn	<string>
service	Name of Service definition	<string>
vlan_id		<integer>
down	Direction the MAC address propagated	<Boolean>
fabric	MAC address is from a fabric device	<Boolean>
org	Vendor Organization name from OUI	<string>



The Slice View query provides a list of Endpoints that are the traffic path between the Endpoint device/port and the WAN connection to the SD-LAN. The list of Endpoints can subsequently be used to retrieve all the detailed information for each endpoint within the slice. The list starts with the requested Endpoint and continues to the TOR(s) endpoints.

Each endpoint is listed with its name and the associated device ID, which can be used with the device_details query.

The following table shows the information returned from the query pointing towards “<site URL>/api/request” either via HTTPS or via POST. It is recommended to use POST so that username/password information is protected.

<https://nnn.nnn.nnn.nnn/api/request?username=admin&password=admin&query=slice&endpoint=xxxxxxxxxx>

FIELD	Definition	Contents
domain	Name of domain	<string>
site	Name of Site	<string>
endpoints	List of endpoints	<array>
	For each entry	
dev_id	Use for detail lookup	<integer>
endpoint_name	Name	<string>
topology_depth	Distance from ToR	<integer>
	Same from previous device	
	...	
	Same for TOR devices	